

Required Minimum Distributions (as a Representation of Actuarial Methods)

One final spending rule serves as a reasonably easy way to implement an actuarial method for retirement spending. Actuarial methods generally have retirees recalculate their sustainable spending annually based on the remaining portfolio balance, remaining longevity, and expected portfolio returns. These are the factors that enter into the PMT formula:

$PMT(rate, nper, pv, fv, type)$.

This formula provides the sustainable spending amount when you have an expected return for investments (*rate*), the planning horizon (*nper*), the current size of the financial portfolio (*pv*), the desired amount of remaining wealth at the end of the planning horizon (*fv*), and a value of 1 for *type* if withdrawals are made at the start of the period. If the *rate* is expressed in inflation-adjusted terms, the answer would imply an inflation-adjusted spending level. In addition, *fv* could be a value less than zero (to reflect a distribution at the end date) if the retiree wants to leave a bequest or to preserve a portion of the portfolio for other purposes.

Though the formula provides an ongoing sustainable spending amount, this calculation could be repeated annually, reflecting ongoing changes in *rate*, *nper*, and *pv*, which would then provide retirees with updated sustainable spending amounts for each year of retirement. The remaining portfolio balance will evolve over time, and circumstances may also call for a change in expected market returns. A dynamic measure of remaining life expectancy is also important, as withdrawal rates can increase as the remaining time horizon shortens. For various actuarial methods, differences center primarily on appropriate assumptions for market returns and remaining longevity, as well as whether additional smoothing factors should be applied to control how quickly spending adjusts to new circumstances.

A simple form for the actuarial method is to use the Internal Revenue Services' required minimum distribution (RMD) rules as a more general guide for sustainable spending. In an effort to get those benefiting from tax deferral to eventually pay taxes, the RMD rules indicate a by-age percentage that must be withdrawn from tax-deferred accounts. Two research articles from 2012 (David Blanchett, Maciej Kowara, and Peng Chen's "Optimal Withdrawal Strategy for Retirement-Income Portfolios" in the *Retirement*